Footy ELO: A Position-Aware Rating System for Europe's Top 5 Leagues

Proposal & Motivation

Football analytics has exploded, yet **traditional end-of-season awards** (Ballon d'Or, PFA Player of the Year) often suffer from:

- **PR & popularity bias** – big names, big clubs.
- **Attacker over-representation** – goals are flashy, defenders & goalkeepers are under-valued.
- **Static, one-off voting** – no dynamic, match-by-match tracking of form.

We propose **Footy ELO** – a **data-driven, position-aware** rating system that:

- Normalises performance **within each position** (GK / DEF / MID / ATT).
- Uses granular per-90 stats (from FotMob) to compute a daily/weekly ELO score.
- Mirrors the ELO logic from chess/competitive gaming, but adapted for football.

> **Note**
>
> **Goal** – A living, unbiased ranking that rewards *actual* contribution, not reputation.

Data Source – FotMob + Reus

All raw stats are pulled from **FotMob** using the open-source scraper **Reus**.

- Player metadata: name, team, position (derived from average pitch position).
- Per-90 stats: goals, assists, shots, passes, tackles, interceptions, saves, cards, etc.
- Season: 2024-25 and 2025-26 (where available).
- So far the prototype website is hosted on localhost only

Reus produces a CSV per league, e.g. `pl.csv`, `laliga.csv`.
Our ELO engine reads these CSVs and outputs `*_elo.csv`.

ELO Mathematics (Adjusted for Football)

We start from a **base ELO** of **1500** and a **K-factor** of **10**.

Step 1 – Raw performance delta

For a player with position (p) and a set of statistical contributions (s_i) (each weighted by $(w_i^{(p)})$), the raw delta is:

$$\Delta_{\text{raw}} = K \cdot \sum_i \left(s_i \cdot w_i^{(p)} \right)$$

where (K = 10) (the ELO_CONFIG["K_FACTOR"]).
Then:

$$\text{ELO}_{\text{raw}} = \text{BASE_ELO} + \Delta_{\text{raw}}$$

Step 2 – Position normalisation

Because attackers naturally have higher raw scores (goals weigh heavily), we normalise **within each position**:

$$\text{ELO}_{\text{final}} = \text{BASE_ELO} + (\text{ELO}_{\text{raw}} - \mu_{\text{pos}})$$

where (\mu_{\text{pos}}) is the **mean raw ELO** of all players in that position.
This centres every position around 1500, making **cross-position comparisons fair**.

Step 3 – Clamping & rounding

- Minimum ELO = 800 (prevents unreasonably low ratings).
- Final value rounded to nearest integer.

> $$\text{ELO}_{\text{raw}} = 1500 + 10 \sum_i w_i^{(p)} \cdot s_i$$
>
> $$\text{ELO}_{\text{final}} = \max\left(800,\ \text{round}\left(1500 + (\text{ELO}_{\text{raw}} - \bar{\text{ELO}}_{\text{pos}})\right)\right)$$

Position-Specific Weights (excerpt)

Each stat's weight depends on the player's position.
Here are the **most influential weights** per role (full list in code).

```
BASE_ELO = 1500

ELO_CONFIG = {
    "K_FACTOR": 10,
    "MIN_ELO": 800,
}

STAT_WEIGHTS = {
"GK": {
    "Top scorer": 1.5,
    "Assists": 1.0,
    "Fouls committed per 90": -0.9,
    "Yellow cards": -1.5,
    "Red cards": -2.0,
    "Saves per 90": 1.5,
    "Goals conceded per 90": -0.9,
    "Goals prevented": 1.2,
    "Clean sheets": 1.5,
    "Penalties conceded": -0.7,
},

"DEF": {
    "Top scorer": 1.5,
    "Assists": 1.0,
    "Fouls committed per 90": -0.9,
    "Yellow cards": -1.5,
    "Red cards": -2.0,
    "Clearances per 90": 1.5,
    "Blocks per 90": 1.5,
    "Interceptions per 90": 3.0,
    "Successful tackles per 90": 2.0,
    "Accurate long balls per 90": 0.2,
    "Clean sheets": 0.9,
},

"MID": {
    "Top scorer": 1.5,
```

```
    "Assists": 1.0,
    "Fouls committed per 90": -0.9,
    "Yellow cards": -1.5,
    "Red cards": -2.0,
    "Accurate passes per 90": 0.1,
    "Accurate long balls per 90": 0.2,
    "Possession won final 3rd per 90": 0.6,
    "Interceptions per 90": 3.0,
    "Expected goals (xG)": 0.2,
    "Expected assist (xA)": 0.5,
},

"ATT": {
    "Top scorer": 1.5,
    "Assists": 1.0,
    "Fouls committed per 90": -0.9,
    "Yellow cards": -1.5,
    "Red cards": -2.0,
    "Goals per 90": 1.6,
    "Shots on target per 90": 1.0,
    "Successful dribbles per 90": 1.2,
    "Big chances missed": -0.8,
}

}
```

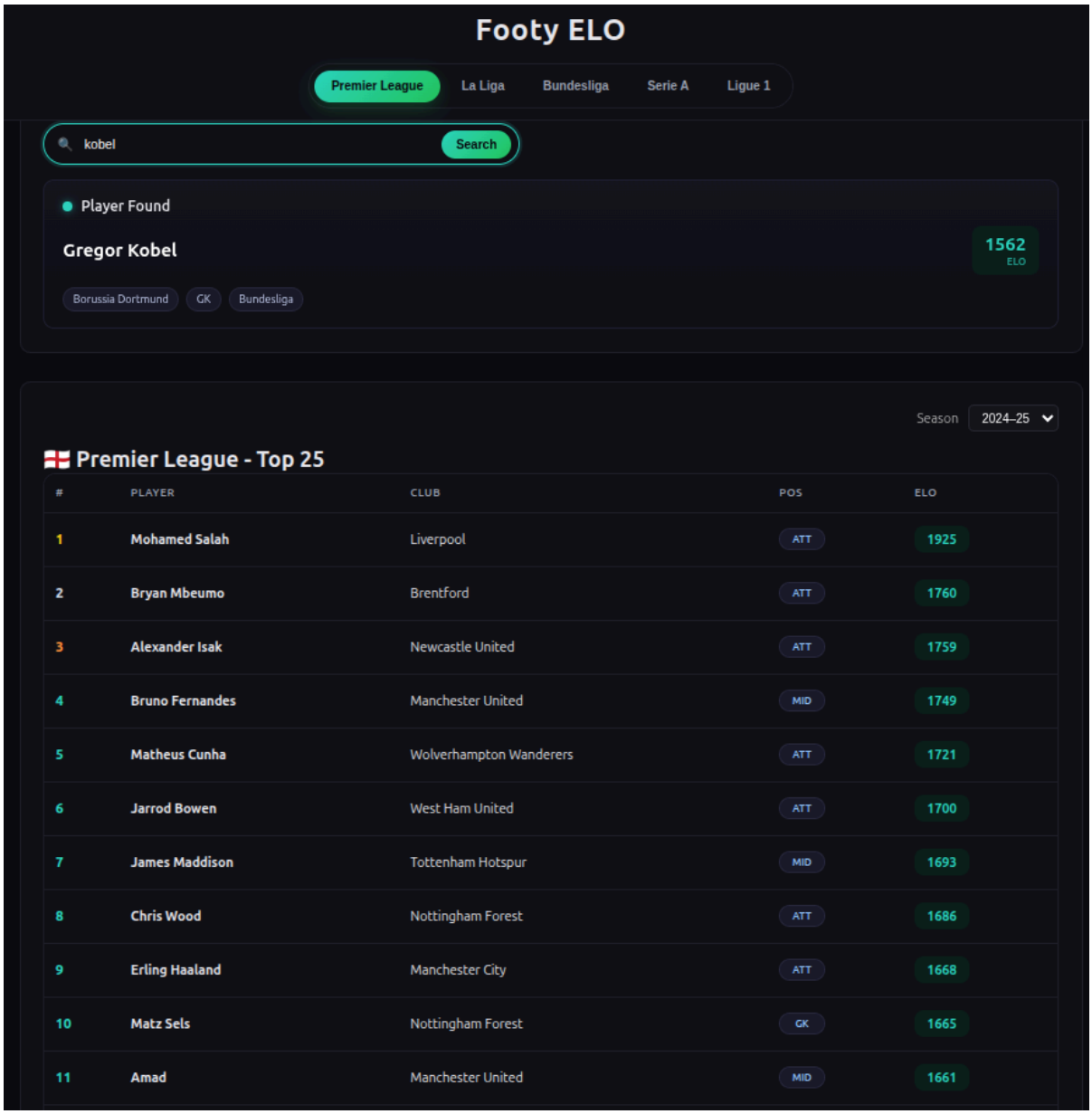
Note

- Negative weights (e.g. fouls, cards, goals conceded) punish poor behaviour.
- Top Scorer is FotMob's way of tracking Goal's Scored

Inference from Current Leaderboards

I have four snapshots (images provided). Below are key observations.

🏴 Premier League 2024-25



- **Attackers dominate** the top, but Bruno Fernandes (MID) breaks in at #4.
- Haaland "only" 1668 – possibly due to injuries or fewer per-90 contributions beyond goals.

🇪🇸 La Liga 2024-25

Season 2024–25

🇪🇸 La Liga - Top 25

#	PLAYER	CLUB	POS	ELO
1	Kylian Mbappé	Real Madrid	ATT	1753
2	Robert Lewandowski	Barcelona	ATT	1745
3	Pedri	Barcelona	MID	1704
4	Ante Budimir	Osasuna	ATT	1701
5	Oihan Sancet	Athletic Club	ATT	1686
6	Julián Álvarez	Atlético de Madrid	ATT	1684
7	Joan Garcia	Espanyol	GK	1672
8	Lamine Yamal	Barcelona	ATT	1670
9	Raphinha	Barcelona	ATT	1669
10	Ayoze Pérez	Villarreal	ATT	1667

- Tighter spread than PL – top score 1753 vs PL's 1925.
- Goalkeeper (Joan García) appears at #7 – unusual but possible with many saves/clean sheets.
- Surprising to see Balondor cotenders Yamal & Raphina not have as great laliga season compared to Champions league

🇮🇹 Serie A 2025-26

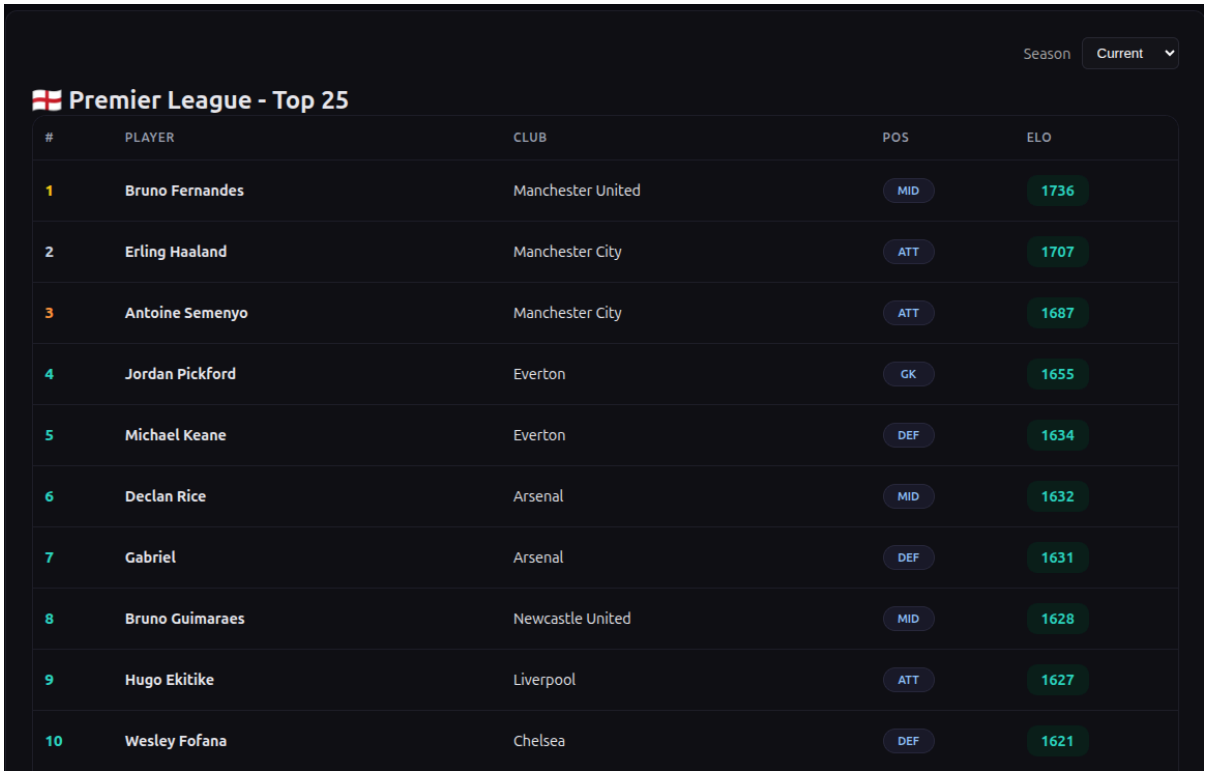
Season Current

🇮🇹 Serie A - Top 25

#	PLAYER	CLUB	POS	ELO
1	Federico Dimarco	Internazionale	MID	1708
2	Ivan Provedel	Lazio	GK	1695
3	Mile Svilar	Roma	GK	1690
4	Marco Carnesecchi	Atalanta	GK	1666
5	Mike Maignan	Milan	GK	1659
6	Hakan Calhanoglu	Internazionale	MID	1646
7	Jean Butez	Como	GK	1646
8	Scott McTominay	Napoli	MID	1637
9	Kenan Yildiz	Juventus	ATT	1635
10	Tarik Muharemovic	Sassuolo	DEF	1627

- **Very high representation of goalkeepers** – likely due to Serie A's defensive style (many saves / low goals conceded).
- Dimarco (attacking full-back classified as MID) leads.

🏴 Premier League 2025-26



Season Current

🏴 Premier League - Top 25

#	PLAYER	CLUB	POS	ELO
1	Bruno Fernandes	Manchester United	MID	1736
2	Erling Haaland	Manchester City	ATT	1707
3	Antoine Semenyo	Manchester City	ATT	1687
4	Jordan Pickford	Everton	GK	1655
5	Michael Keane	Everton	DEF	1634
6	Declan Rice	Arsenal	MID	1632
7	Gabriel	Arsenal	DEF	1631
8	Bruno Guimaraes	Newcastle United	MID	1628
9	Hugo Ekitike	Liverpool	ATT	1627
10	Wesley Fofana	Chelsea	DEF	1621

- **Midfielder at #1** – Bruno Fernandes overtakes Haaland. This is as of January , When Micheal Carrik took over , and Haaland was ghosting most of the matches
- Only one defender in top 10 (Fofana), reflecting the system's slight attacker/mid bias but far less than traditional awards.

Doubts & Potential Improvements

Current Strengths

- **Position normalisation** works – we see GKs in Serie A top 5, which never happens in raw goal based rankings.
- **Negative weights** penalise cards & fouls, encouraging fair play.

Doubts / Open Questions

1. **No opponent strength adjustment**
   - Classic ELO updates after matches using opponent's rating. Our system is purely stat-based.
   - *Improvement*: Add a match-level correction:

$$[\Delta_{\text{match}} = K \cdot (S_{\text{actual}} - S_{\text{expected}})]$$

where (S_{expected}) depends on opponent's average ELO.

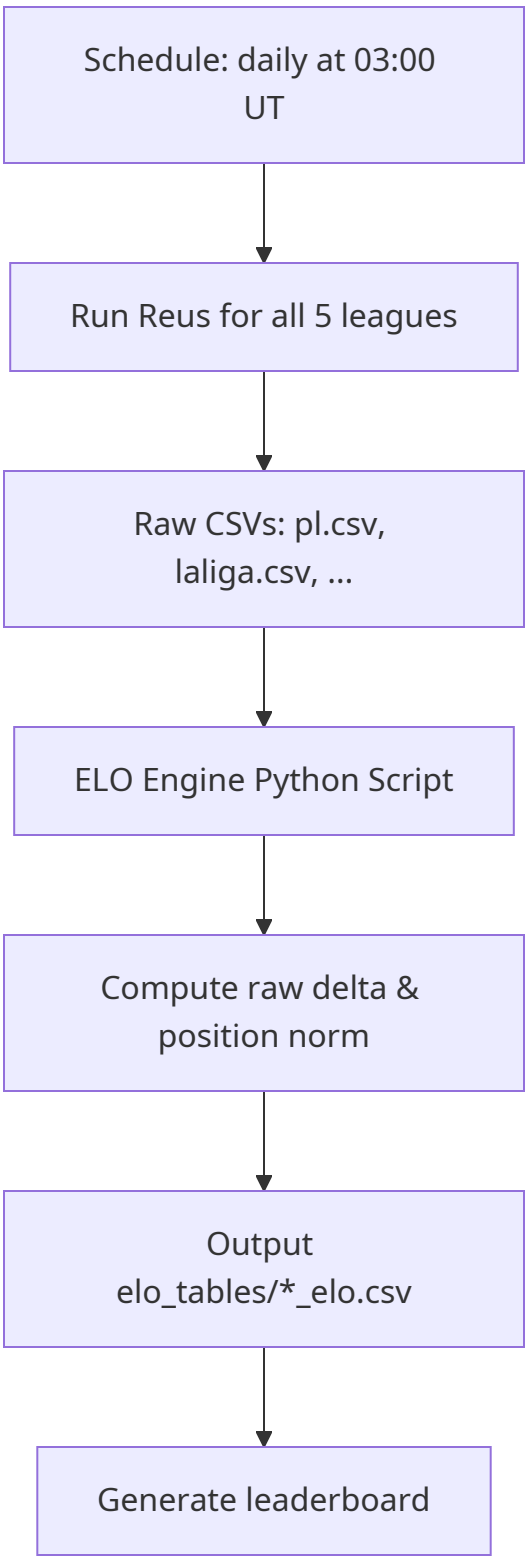
2. **Static weights**
   - Weights are fixed (e.g. `Interceptions = 3.0` for DEF). But the value of an interception varies by game state.
   - *Improvement*: Use machine learning (e.g. Ridge regression) to learn weights from past award results.
3. **No time decay**
   - A great performance from 3 months ago still counts equally.
   - *Improvement*: Apply exponential decay:

     $$[\text{ELO}_{\text{decayed}} = \text{ELO}_{\text{old}} \cdot e^{-\lambda \cdot \text{days}}]$$

     with $(\lambda = 0.005)$ (≈ 5% loss per 10 days).
4. **Position classification**
   - "Average pitch position" from FotMob is good but can be noisy (e.g. wing-backs sometimes labelled DEF or MID).
   - *Improvement*: Cluster players using actual heatmaps or use a consensus from multiple sources (Transfermarkt, WhoScored).
5. **League strength normalization**
   - A POT of 90 in the Premier League ≠ 90 in the Belgian league.
   - *Improvement*: Multiply raw POT by a **league coefficient** (e.g., EPL = 1.2, Eredivisie = 0.85), or recenter ELO means per league.
6. **Competition weighting**
   - World Cup matches should matter more than friendlies; UCL knockouts more than group stage.
   - *Improvement*: Assign multipliers:
     - World Cup / UCL final = 3.0×
     - UCL knockout / Euros = 2.0×
     - League matches = 1.0×
     - Friendlies = 0.3× (or exclude)
7. **Annual rating (best player over a calendar year)**
   - Simply summing ELO changes rewards high match volume, not quality.
   - *Improvement*: Use **weighted average** of competition-adjusted ELOs, weighted by minutes played, then decay older months exponentially.

Pipeline Automation (Critical Improvement)

Currently the process is semi-manual:



- Store historical ELO in a SQLite database to track **ELO over time** (form curves).
- Add **unit tests** to catch weight changes that break normalisation.

Also would love your opinion on the pipeline to make it more scaleable also ?

Conclusion

The **Footy ELO** system gives every player – from goalkeeper to striker – a **fair, dynamic rating** based on what they *actually do* on the pitch.
By normalising per position and using detailed per-90 stats, we reduce the classic "attacker bias".

The next step is **automating the pipeline** (Reus → ELO → dashboard) and **introducing opponent strength & time decay** for a true ELO experience.

> ✓ **Finally !**
>
> *"ELO doesn't lie – stats do."*
> *(with the right weights, of course)*